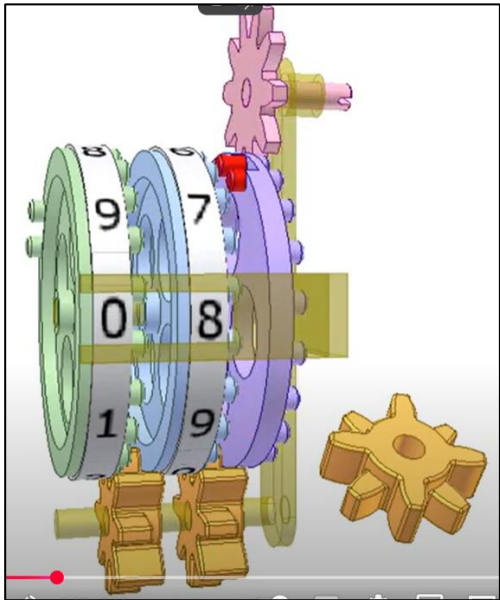
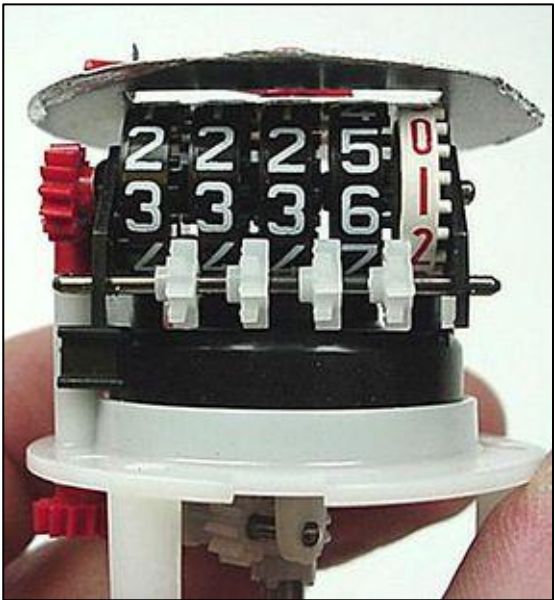
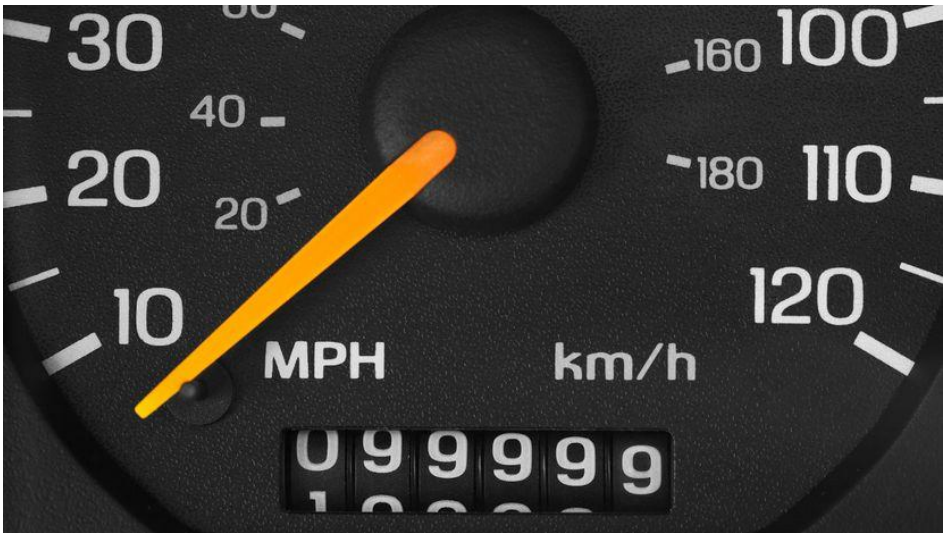


counting with
10 symbols / characters / digits

also known as
“**decimal**” counting

decade
decagon
decem (latin)
deca (greek)
dieci (italian)
diez (spanish)
dix (french)
december (is it 12 ? or 10 ?!)

why 10 ? people like counting with 10 symbols / digits
because we have 10 fingers (10 digits)



[video of mechanical decimal counter](#)

counting with
2 symbols

also known as
“**binary**” counting

bicycle
biped
bi

why 2 ? computers have various ways of representing individual values, each with only 2 possible states

voltage on wire	high voltage / low voltage
electronic switch	on / off
CD surface state	hole / no hole
piece of magnetic tape	magnetized / not magnetized
boolean value	true / false



[video of mechanical binary counter](#)

	10 ²	10 ¹	10 ⁰		2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		16 ²	16 ¹	16 ⁰
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0	0	0	0	1	1	0	0	1
2	0	0	2	0	0	0	0	0	0	1	0	0	0	0	2
3	0	0	3	1	0	0	0	0	0	1	1	1	0	0	3
4	0	0	4	0	0	0	0	0	1	0	0	0	0	0	4
5	0	0	5	0	0	0	0	0	1	0	1	0	0	0	5
6	0	0	6	0	0	0	0	0	1	1	0	0	0	0	6
7	0	0	7	0	0	0	0	0	1	1	1	1	0	0	7
8	0	0	8	0	0	0	0	1	0	0	0	0	0	0	8
9	0	0	9	0	0	0	0	1	0	0	1	0	0	0	9
10	0	1	0	0	0	0	0	1	0	1	0	0	0	1	A
11	0	1	1	0	0	0	0	1	0	1	1	0	0	1	B
12	0	1	2	1	0	0	0	1	1	0	0	0	0	2	C
13	0	1	3	0	0	0	0	1	1	0	1	0	0	3	D
14	0	1	4	1	0	0	0	1	1	1	0	0	0	4	E
15	0	1	5	0	0	0	0	1	1	1	1	0	0	5	F
16	0	1	6	1	0	0	1	0	0	0	0	0	1	0	10
17	0	1	7	0	0	0	1	0	0	0	1	0	0	1	11
18	0	1	8	1	0	0	1	0	0	1	0	0	0	2	12
19	0	1	9	0	0	0	1	0	0	1	1	0	0	3	13
20	0	2	0	1	0	0	1	0	1	0	0	0	0	4	14
21	0	2	1	0	0	0	1	0	1	0	1	0	0	5	15
22	0	2	2	1	0	0	1	0	1	1	0	0	1	6	16
23	0	2	3	0	0	0	1	0	1	1	1	0	0	7	17
24	0	2	4	1	0	0	1	1	0	0	0	0	0	8	18
25	0	2	5	0	0	0	1	1	0	0	1	0	0	9	19
26	0	2	6	1	0	0	1	1	0	1	0	0	0	A	20
27	0	2	7	0	0	0	1	1	0	1	1	0	0	B	21
28	0	2	8	1	0	0	1	1	1	0	0	0	0	C	22
29	0	2	9	0	0	0	1	1	1	1	0	0	0	D	23
30	0	3	0	1	0	0	1	1	1	1	1	0	0	E	24
31	0	3	1	0	0	0	1	1	1	1	1	1	0	F	25
32	0	3	2	1	0	1	0	0	0	0	0	0	0	10	26
33	0	3	3	0	0	1	0	0	0	0	0	1	0	11	27
34	0	3	4	1	0	1	0	0	0	1	0	0	0	12	28



p41. "Hexadecimal uses sixteen different **digits**."
 It would be clearer to say :
 "Hexadecimal uses sixteen different symbols."

p42. "Each hex **digit** is equal to a nibble in binary"
 It would be clearer to say :
 "Each hex character represents four binary characters, also known as a nibble."

The cycling distance from Beacon Croft to Ironbridge is :
 32 miles (decimal) "thirty two miles"
 100000 miles (binary) "binary one zero zero zero zero zero miles"
 20 miles (hexadecimal) "hex two zero miles"

a given symbol in hexadecimal **does** always line up with the same symbols in binary

a given symbol in decimal **doesn't** always line up with the same symbols in binary

Finish

	10^2	10^1	10^0
0	0	0	0
1	0	0	1
2	0	0	2
3	0	0	3
4	0	0	4
5	0	0	5
6	0	0	6
7	0	0	7
8	0	0	8
9	0	0	9
0	1	0	
0	1	1	
0	1	2	
0	1	3	
0	1	4	
0	1	5	
0	1	6	
0	1	7	
0	1	8	

...		
2	4	3
2	4	4
2	4	5
2	4	6
2	4	7
2	4	8
2	4	9
2	5	0
2	5	1
2	5	2
2	5	3
2	5	4
2	5	5
2	5	6
2	5	7
2	5	8

2	4	4
---	---	---

	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	1
0	0	0	0	0	0	1	0	1	0
0	0	0	0	0	0	1	1	0	0
0	0	0	0	0	0	1	1	0	1
0	0	0	0	0	0	1	1	1	1
0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	1	0

...								
0	1	1	1	1	0	0	1	1
0	1	1	1	1	0	1	0	0
0	1	1	1	1	0	1	0	1
0	1	1	1	1	0	1	1	0
0	1	1	1	1	0	1	1	1
0	1	1	1	1	1	0	0	0
0	1	1	1	1	1	0	0	1
0	1	1	1	1	1	0	1	0
0	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1
1	0	0	0	0	0	0	1	0

250 shifted left

$$128 + 64 + 32 + 16 + 4 = 244$$

F	4
---	---

	16^2	16^1	16^0
0	0	0	0
1	0	0	1
2	0	0	2
3	0	0	3
4	0	0	4
5	0	0	5
6	0	0	6
7	0	0	7
8	0	0	8
9	0	0	9
A	0	0	A
B	0	0	B
C	0	0	C
D	0	0	D
E	0	0	E
F	0	0	F
0	1	0	
0	1	1	
0	1	2	

...		
0	F	3
0	F	4
0	F	5
0	F	6
0	F	7
0	F	8
0	F	9
0	F	A
0	F	B
0	F	C
0	F	D
0	F	E
0	F	F
1	0	0
1	0	1
1	0	3

0	F	4
---	---	---